E-VOTING SYSTEM

Team Members

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Problem Statement

The traditional voting system involves manual processes that cosume a lot of time and effort. It may lead to delays in vote counting, human errors and chances of duplicate or invalid voting. Managing voter records and announcing results can also be difficult and slow.Therefore, there is a need for an efficient , accurate and secure E-Voting System that simplifies the voting process and provides quick results.

Objective

The main objective of this project is to develop an efficient and user-friendly E-Voting System using C++.It is designed to provide a simple voting process where each voter can cast only one vote. The system maintains voter and candidate records using file handling techniques. It also ensures accurate vote counting and displays results quickly. This project aims to reduce manual effort, save time, and improve the reliability of the voting process.

System Features

Secure Login

ID-based authentication against pre-loaded voter data to ensure only registered users participate.

Automated Tally

Computational logic increments candidate scores instantly, removing human counting errors.

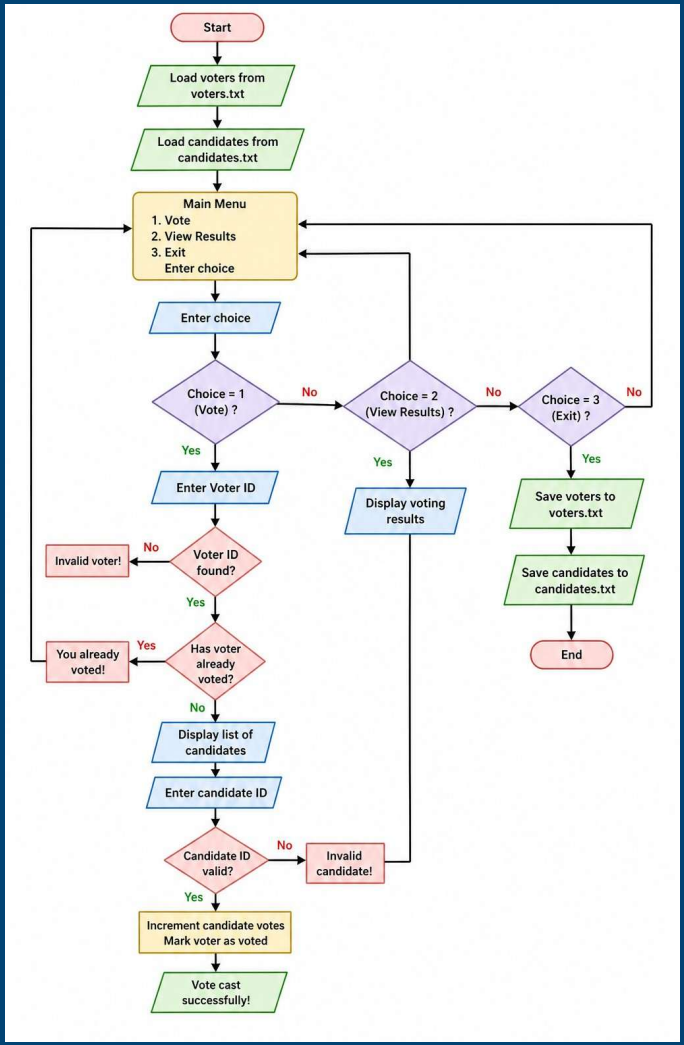
Anti-Fraud Logic

Internal Flags mark a voter as “voted” instantly, blocking any second attempt with the same ID.

File Management

Records are stored in external text files(voters.txt and candidates.txt) for auditabality.

Flow Chart


Start
Load voters from voters.txt
Load candidates from candidates.txt
Main Menu
1. Vote
2. View Results
3. Exit
Enter choice
Enter choice
Choice = 1 (Vote) ?
No
Choice = 2 (View Results) ?
No
Choice = 3 (Exit) ?
No
Yes
Yes
Yes
Enter Voter ID
Display voting results
Save voters to voters.txt
Save candidates to candidates.txt
End
Voter ID found?
No
Invalid voter!
Yes
Has voter already voted?
Yes
You already voted!
No
Display list of candidates
Enter candidate ID
Candidate ID valid?
No
Invalid candidate!
Yes
Increment candidate votes
Mark voter as voted
Vote cast successfully!

Implementation

Object-Oriented Programming

The system is built using classes and objects to manage voter and candidate data as encapsulated entitles.

File I/O Mechanism

Uses ifstream to load data at startup and ofstream to write final results back to the database files.

Control Structures

Implementation of switch-case for the main menu and conditional loops for secure input validation.

Core Module

Module Name	File Operation
• Data Initialization	Read (voters.txt)
• Authentication	Search
• Voting Logic	Update
• Data Persistance	Write (candidates.txt)

File Handling Mechanism

Voter Database

The voters.txt file stores unique IDs followed by a boolean flag(0 for not voted, 1 for voted).

Candidate Records

The candidates.txt file maintains the names of candidates and their cumulative vote counts.

System reads all data into memory at startup to minimize disk access during the voting phase.

Testing Outcomes

Standard User Flow

Passed: Valid ID entry --> Selection --> Success message --> Marked as Voted.

Security Testing

Passed: Entering a previosly used ID triggers “Already Voted” alert and block access.

Boundary Testing

Passed: Entering an ID not in the file triggers “Invalid ID: error and return to menu.

Conclusion

Project Summary

The E-voting system successfully demonstrates the transition from manual to digital voting using c++. It provides a secure, efficient, and reliable method for conducting elections.

Final Conclusion

By leveraging OOP and File Handling, we have addressed the key problems of human error and data loss, ensuring a transport and accurate democratic process.

THANK YOU